# WEEK 03
# Exploratory Data Analysis (EDA) & Visualization

Python · Pandas · Matplotlib · Seaborn

## Purpose and scope

This guide teaches concepts and worked examples for descriptive statistics and visualization. Nine separate dataset-specific workbooks and executable notebooks are planned as later deliverables; this guide does not replace them.

Curriculum requirements: explain EDA; summarize numerical and categorical data with mean, median, mode, standard deviation and counts; build histograms, box plots, scatter plots, bar charts and heatmaps; identify patterns, outliers and relationships; document findings; submit an EDA report with at least five plot types and complete peer review. The daily sequence and additional examples here are instructional extensions.

## Learning route

| Day | Concepts | Proof of competence |
|---|---|---|
| 1 | Question framing, grain, descriptive statistics | Correct summaries with denominators |
| 2 | Univariate analysis | Histogram, box plot, categorical bar plot |
| 3 | Relationships and group comparisons | Scatter plot, heatmap, defensible comparisons |
| 4 | Outliers, ethics and documentation | Sensitivity check and documented EDA |
| 5 | Independent investigation and peer review | Reproducible notebook, five plot types, review |

# Chapter 1 — What EDA is and is not

Exploratory Data Analysis is a structured process for understanding the distribution, completeness, variation and relationships of variables before drawing conclusions or building models. It is iterative: a chart may reveal an issue that sends you back to verify data definitions. It is not merely producing charts.

| Stage | Question | Typical evidence |
|---|---|---|
| **Frame** | What decision needs evidence? | Business question, scope, grain |
| **Profile** | What data do we have? | Rows, types, missingness, uniqueness |
| **Describe** | What is typical and variable? | Summary tables, distributions |
| **Compare** | How do groups differ? | Grouped statistics, bar/box plots |
| **Investigate** | Are patterns robust? | Outlier and subgroup checks |
| **Communicate** | What is supported? | Captions, limitations, reproducible report |

## 1.1 The unit of analysis (“grain”)

A line-item table contains one record per product on an invoice. An invoice-level table contains one record per order. Averaging line-item revenue does not give average order value. State grain before selecting a denominator.

#### Stop & predict

A customer buys three different products in one invoice. How many orders are there?

Self-check: One invoice/order, but three line items.

## 1.2 Observation, association, causation

Observation: “High-spending customers appear more frequently in the sample.” Association: “Spending and visit frequency rise together.” Causation: “More visits cause spending to rise.” EDA generally supports the first two, not the third. Consider confounding, selection bias, reverse causation and measurement error.

# Chapter 2 — Set up a reproducible practice dataset

The following 12-row, fictional Nigerian retail dataset is deliberately small and includes an extreme order value, an unknown age, two cities and two channels. It is designed for teaching; do not interpret its statistics as evidence about actual customers.

```
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
orders = pd.DataFrame({
    "order_id": range(101, 113),
    "city": ["Lagos","Abuja","Lagos","Kano","Abuja","Lagos",
             "Kano","Abuja","Lagos","Kano","Abuja","Lagos"],
    "channel": ["Online","Store","Online","Store","Online","Store",
                "Online","Online","Store","Store","Online","Online"],
    "order_value": [2000,4000,3000,3500,5000,4500,
                    2500,3000,80000,4000,5500,3500],
    "delivery_days": [2,4,3,5,2,3,6,2,4,5,3,2],
    "age": [25,34,29,41,None,31,28,35,52,46,33,27]
})
assert orders["order_id"].is_unique
print(orders.shape)    # (12, 6)
print(orders.head())
print(orders.dtypes)
```

Run this cell first in a new notebook. All subsequent code assumes the variable orders exists. When adapting to a public dataset, replace orders with your Week 2 cleaned DataFrame and inspect actual column names.

## 2.1 A data-quality checkpoint

```
print(orders.info())
print(orders.isna().sum())
print(orders.nunique(dropna=False))
print(orders.duplicated(subset=["order_id"]).sum())
print(orders["city"].value_counts(dropna=False))
```

Expected: 12 rows; one missing age; no duplicate order IDs; Lagos 5 orders, Abuja 4, Kano 3. The missing age does not prevent analysis of order_value, but does affect age-based summaries.

#### Stop & predict

If age is missing for one order, should the whole order be removed before analyzing revenue?

Self-check: Not automatically. Missingness decisions depend on the analytical question and variables required.

# Chapter 3 — Descriptive statistics with Pandas

## 3.1 Count, missingness, mode and frequency

```
print(orders["age"].count())        # 11 non-null
print(orders["age"].isna().sum())   # 1 missing
print(orders["city"].mode())
print(orders["city"].value_counts(dropna=False))
print(orders["channel"].value_counts(normalize=True).mul(100))
```

Use count() for non-null observations, size/len for all rows, value_counts() for frequencies, and normalize=True for proportions. Always identify the denominator; 11 observed ages do not mean there are only 11 orders.

## 3.2 Mean, median, standard deviation and quartiles

```
x = orders["order_value"]
print(x.mean())       # 10000.0
print(x.median())     # 3750.0
print(x.mode())       # 3000, 3500, 4000 (multimodal)
print(x.std())        # sample standard deviation
print(x.quantile([.25,.5,.75,.95]))
print(x.describe())
```

The mean (10,000) is far above the median (3,750) because of the 80,000 order. Standard deviation measures dispersion in the same units as the variable. Pandas std() uses sample standard deviation (ddof=1) by default.

## 3.3 Grouped statistics

```
summary = orders.groupby("city", as_index=False).agg(
    orders=("order_id","nunique"),
    total_revenue=("order_value","sum"),
    mean_order=("order_value","mean"),
    median_order=("order_value","median")
)
print(summary.sort_values("total_revenue", ascending=False))
print(orders.groupby("channel")["order_value"].describe())
```

Lagos contains the 80,000 order; its mean may not describe a typical Lagos purchase. Compare mean and median within each group. Aggregated revenue and mean order answer different management questions.

### Practice 3A — A management summary

Your task: Report orders, total revenue, mean and median order value for each city. Identify the city with highest revenue and explain whether that proves the city has the highest typical order value.

Evidence to record: Include the table, denominator and a two-sentence interpretation.

### Practice 3B — Missing age

Your task: Compute average age by channel and report the number of non-null ages in each group. Explain why reporting means without counts could mislead.

Evidence to record: Submit group mean and count side-by-side.

## Chapter 4 — Visualization principles

A good chart answers a specific question, matches variable types, states units, uses an honest scale, and includes a brief interpretation. Titles should state the metric, population and (when relevant) period. Use figure sizing, tight_layout(), labels and readable categories.

| Question / variables | Suitable plot | Watch for |
|---|---|---|
| **Numeric distribution** | Histogram | Bin width and skew |
| **Numeric spread / extreme values** | Box plot | Outliers are not automatically errors |
| **Categorical counts or group totals** | Bar plot | Correct measure and zero baseline |
| **Two numerical variables** | Scatter plot | Overplotting and causality |
| **Correlation matrix or category cross-tab** | Heatmap | Meaning of cells, scale, missingness |

# Chapter 5 — Histogram: understand a distribution

```
fig, ax = plt.subplots(figsize=(8, 4))
sns.histplot(data=orders, x="order_value", bins=8, ax=ax)
ax.set(title="Order value distribution — fictional retail orders",
       xlabel="Order value (NGN)", ylabel="Number of orders")
plt.tight_layout()
plt.show()
```

The chart will show most values between 2,000 and 5,500, plus an isolated value at 80,000. Because the dataset is small, bin selection strongly affects appearance; inspect the underlying numbers rather than overinterpreting bin shape.

```
fig, ax = plt.subplots(figsize=(8,4))
sns.histplot(data=orders, x="order_value", bins=[0,2000,4000,6000,10000,100000],
             ax=ax)
ax.set(xlabel="Order value (NGN)", ylabel="Orders",
       title="Order values with business-defined bins")
plt.tight_layout(); plt.show()
```

### Stop & predict

What changes when bins change?

Self-check: The visual emphasis changes; the underlying observations do not.

### Practice 5A — Compare bin choices

Your task: Create two histograms of order_value using different bins. Which presentation makes the extreme value easiest to notice?

Evidence to record: Record both figures and explain why bin choice matters.

# Chapter 6 — Box plot and outlier interpretation

```
fig, ax = plt.subplots(figsize=(8,3))
sns.boxplot(data=orders, x="order_value", ax=ax)
ax.set(title="Order value spread and potential outliers",
       xlabel="Order value (NGN)")
plt.tight_layout(); plt.show()

q1, q3 = orders["order_value"].quantile([.25,.75])
iqr = q3 - q1
lower, upper = q1 - 1.5*iqr, q3 + 1.5*iqr
flagged = orders.loc[
    (orders["order_value"] < lower) |
    (orders["order_value"] > upper)
]
print(q1, q3, lower, upper)
print(flagged[["order_id","order_value"]])
```

The IQR rule flags unusual values relative to the middle 50% of data. It does not determine whether a record is erroneous. Verify source transactions before deleting or winsorizing. If the 80,000 order is real, it may be commercially important.

```
print("Mean all:", orders["order_value"].mean())
print("Mean without flagged:",
      orders.loc[~orders.index.isin(flagged.index),"order_value"].mean())
print("Median all:", orders["order_value"].median())
```

### Practice 6A — Sensitivity analysis

Your task: Report mean and median with and without the flagged order(s). Write whether excluding them is defensible and what additional source information you would need.

Evidence to record: A before/after table and an explicit decision not to silently delete valid extremes.

# Chapter 7 — Bar charts and categorical comparison

```
counts = orders["city"].value_counts().rename_axis("city").reset_index(name="orders")
fig, ax = plt.subplots(figsize=(7,4))
sns.barplot(data=counts, x="city", y="orders", ax=ax)
ax.set(title="Order count by city", xlabel="City", ylabel="Number of orders")
ax.set_ylim(bottom=0)
plt.tight_layout(); plt.show()

revenue = orders.groupby("city", as_index=False)["order_value"].sum()
fig, ax = plt.subplots(figsize=(7,4))
sns.barplot(data=revenue, x="city", y="order_value", ax=ax)
ax.set(title="Total revenue by city", xlabel="City", ylabel="Revenue (NGN)")
ax.set_ylim(bottom=0)
plt.tight_layout(); plt.show()
```

These plots can produce different rankings. A city with fewer orders can have more revenue because order values differ. Never title a frequency chart “sales” without specifying whether sales means orders, items or money.

### Practice 7A — Choosing a business metric

Your task: A manager asks “Which city performs best?” Create separate charts for order count, total revenue and median order value. Explain why the question is ambiguous.

Evidence to record: Three labeled plots and a proposed definition of performance.

# Chapter 8 — Scatter plots and relationships

```
fig, ax = plt.subplots(figsize=(7,4))
sns.scatterplot(data=orders, x="delivery_days", y="order_value",
                hue="channel", s=85, ax=ax)
ax.set(title="Delivery time versus order value",
       xlabel="Delivery time (days)", ylabel="Order value (NGN)")
plt.tight_layout(); plt.show()

print(orders[["delivery_days","order_value"]].corr(method="pearson"))
print(orders[["delivery_days","order_value"]].corr(method="spearman"))
```

Pearson measures linear association; Spearman measures monotonic rank association. An outlier may change Pearson substantially. Neither measure proves that delivery time affects order value. Consider whether order value is recorded before delivery and whether location/channel influences both.

### Stop & predict

Would a correlation of +0.8 prove that slow deliveries cause high spending?

Self-check: No. Association alone does not establish causation.

### Practice 8A — Investigate a possible relationship

Your task: Plot age versus order_value after excluding missing ages only for this plot. Compute Pearson and Spearman correlations. Explain sample size and why a small sample limits inference.

Evidence to record: Plot, two coefficients and an interpretation that avoids causal claims.

# Chapter 9 — Heatmaps and multivariate summaries

```
numeric = orders[["order_value","delivery_days","age"]]
corr = numeric.corr(numeric_only=True)
fig, ax = plt.subplots(figsize=(6,4))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm",
            vmin=-1, vmax=1, center=0, ax=ax)
ax.set(title="Correlation among numeric variables")
plt.tight_layout(); plt.show()
```

A correlation heatmap uses pairwise non-missing observations by default; the sample size can differ between pairs. Strong colors are not evidence of statistical or practical significance, especially in tiny datasets.

```
cross = pd.crosstab(orders["city"], orders["channel"])
fig, ax = plt.subplots(figsize=(6,4))
sns.heatmap(cross, annot=True, fmt="d", cmap="Blues", ax=ax)
ax.set(title="Order counts: city by channel",
       xlabel="Sales channel", ylabel="City")
plt.tight_layout(); plt.show()
```

This second heatmap shows counts rather than correlations. The color scale therefore has a different meaning. For fair comparison across cities of unequal sizes, consider row-normalized percentages.

```
pct = pd.crosstab(orders["city"], orders["channel"],
                  normalize="index").mul(100)
print(pct.round(1))
```

### Practice 9A — Heatmap denominators

Your task: Compare the count and row-percentage city-by-channel tables. Which answers “Where do most online orders originate?” and which answers “What share of each city’s orders is online?”

Evidence to record: Both heatmaps or tables, with correctly distinguished denominators.

## Chapter 10 — Grouped visual comparisons

```
fig, ax = plt.subplots(figsize=(7,4))
sns.boxplot(data=orders, x="city", y="order_value", ax=ax)
ax.set(title="Order value distribution by city",
       xlabel="City", ylabel="Order value (NGN)")
plt.tight_layout(); plt.show()

city_summary = orders.groupby("city")["order_value"].agg(
    ["count","mean","median","std"])
print(city_summary)
```

Always report group sample sizes. With only 3–5 orders per city, box plots are fragile summaries; individual data points may be more informative. The real datasets later in the programme allow more meaningful group comparisons.

## Chapter 11 — Common statistical and visualization mistakes

| Mistake | Why it matters | Better practice |
|---|---|---|
| **Average line items to estimate order value** | Wrong grain | Aggregate to one row per invoice first |
| **Compare groups without sample sizes** | Unstable small groups | Show n alongside statistics |
| **Treat missing values as zero** | Changes meaning | State a defensible missing-data policy |
| **Delete all box-plot outliers** | Can discard real events | Investigate and compare sensitivity |
| **Infer cause from scatter plot** | Confounding and reverse causality | Use association language |
| **Use a bar chart for continuous distribution** | Obscures numeric shape | Use histogram |
| **Truncate bar-axis baseline** | Exaggerates group differences | Start at zero for magnitude bars |
| **Forget chart units and titles** | Audience cannot interpret | Label population, metric and units |
| **Use too many variables in one heatmap** | Unreadable and misleading | Select relevant variables |
| **Depend on notebook hidden state** | Not reproducible | Restart kernel and Run All |

## Chapter 12 — Reproducibility, accessibility and ethics

- Keep original files unchanged and identify the exact processed input used for EDA.
- Use relative paths and list package dependencies in README or requirements.txt.
- Record dataset source, extraction date, known limitations and missingness.
- Use descriptive chart titles, axis labels, units and written captions; do not rely on color alone.
- Do not expose personal identifiers in screenshots, notebooks or publicly shared repositories.
- Differentiate genuine evidence from hypotheses and recommendations.

## Chapter 13 — Structure of a complete EDA notebook

1. Title and business context: what decision is being informed?
2. Dataset provenance, time coverage, units and observation grain.
3. Three to five answerable questions and hypotheses (clearly labeled as hypotheses).
4. Import and reproducibility cell; data-quality profile and missingness notes.
5. Descriptive tables for numerical and categorical variables.
6. At least five distinct plot types: histogram, box, scatter, bar and heatmap.
7. A caption under each plot: observation, interpretation and limitation.
8. Outlier/anomaly investigation and sensitivity checks.
9. Synthesis: findings supported by evidence, limitations and next steps.
10. References, peer-review notes and changes made after review.

## Chapter 14 — Five-day trainee practice plan

| Day | Practice | Submit |
|---|---|---|
| **Monday** | Frame questions; calculate descriptive statistics | Question plan and statistical table |
| **Tuesday** | Create histogram, box and bar plots | Three plots with captions |
| **Wednesday** | Create scatter and heatmap; test associations | Two plots and limitations |
| **Thursday** | Investigate anomalies; assemble notebook | Draft notebook and evidence log |
| **Friday** | Run All; peer review; oral defence | Final report and peer-review record |

## Chapter 15 — Independent practice: three difficulty levels

### Basic — Explain a distribution

Your task: Select one numerical variable from a Week 2 cleaned dataset. Calculate count, mean, median, std and quartiles; build a histogram and box plot.

Evidence to record: Explain whether the distribution is symmetric or skewed and cite numeric evidence.

### Intermediate — Compare groups responsibly

Your task: Select a meaningful category (e.g., country, school, campaign segment). Report counts, group means and medians; create a bar plot and grouped box plot.

Evidence to record: State whether sample-size imbalance or missingness affects interpretation.

### Challenge — Test a plausible claim

Your task: Choose two numerical variables, draw a scatter plot, compute Pearson and Spearman correlations, identify outliers and repeat the analysis without flagged extremes (without deleting source rows).

Evidence to record: Compare coefficients, discuss confounding and defend which conclusion is warranted.

## Chapter 16 — Peer review and oral defence

Exchange notebooks with a partner. Run the notebook from a fresh kernel and inspect whether it executes without hidden dependencies. Complete the checklist below and provide one commendation, two actionable improvements and one question.

| Peer-review checkpoint | Yes / No / Notes |
|---|---|
| Question and grain are clear | |
| Source, units and limitations are recorded | |
| Summary statistics use correct denominators | |
| At least five plot types are present | |
| Every figure has labels and an interpretation | |
| Outliers are investigated rather than silently deleted | |
| Claims do not confuse association and causation | |
| Notebook runs from a fresh kernel | |

- Why did you choose a histogram instead of a bar chart?
- What would change if you used median instead of mean?
- Which observation most influences your summary and why?
- How would you verify an apparent correlation?
- What is the most important limitation of your dataset?
- Which peer-review suggestion changed your analysis?

## Chapter 17 — Submission requirements and self-check

Submit one reproducible Jupyter Notebook (or Markdown report with supporting reproducible code) containing descriptive statistics, at least five different plot types and written interpretations. Also submit peer-review evidence. This follows the Week 3 curriculum; the folder structure below is recommended.

```
week03-eda/
  data/processed/         # source or documented access instructions
  notebooks/week03_eda.ipynb
  figures/                # optional exported figures
  reports/peer_review.md
  README.md
  requirements.txt
```

- I can explain the observation grain and the meaning of each statistic.
- I have created five different plot types and interpreted each one.
- I have investigated at least one anomaly or outlier.
- I have identified limitations and avoided unsupported causal claims.
- I have restarted the kernel and successfully run every cell.
- I have completed and responded to a peer review.

## Appendix A — Pandas/Matplotlib/Seaborn quick reference

| Goal | Syntax |
|---|---|
| **Data profile** | df.info(); df.describe(include="all") |
| **Missing values** | df.isna().sum() |
| **Counts / proportions** | s.value_counts(dropna=False); s.value_counts(normalize=True) |
| **Quantiles / dispersion** | s.quantile([.25,.5,.75]); s.std() |
| **Group summaries** | df.groupby("group").agg(n=("id","size"), mean=("x","mean")) |
| **Histogram** | sns.histplot(data=df, x="x", bins=30) |
| **Box plot** | sns.boxplot(data=df, x="group", y="x") |
| **Bar chart** | sns.barplot(data=summary, x="group", y="metric") |
| **Scatter** | sns.scatterplot(data=df, x="x", y="y", hue="group") |
| **Heatmap** | sns.heatmap(df[cols].corr(), annot=True) |
| **Crosstab** | pd.crosstab(df["a"], df["b"], normalize="index") |
| **Save figure** | plt.savefig("figures/chart.png", dpi=150, bbox_inches="tight") |